

Лабораторная работа №3

Агентное моделирование

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Содержание I

Докладчик

- Софич Андрей Геннадьевич



- Провести ознакомление с агентным моделированием

Изучить в теории и на практике возможности агентного моделирования.

Выполнение лабораторной работы

Создаем рабочий каталог и инициализируем проект в julia

```
julia> using Pkg

julia> Pkg.add("DrWatson")
  Updating registry at 'C:\Users\Andrew\.julia\registries\General.toml'
  Resolving package versions...
  Installed Xorg_libpciaccess_jll - v0.18.1+0
  Installed libva_jll - v2.23.0+0
  Installed libdrm_jll - v2.4.125+1
  Project No packages added to or removed from 'C:\Users\Andrew\.julia\environments\v1.12\Project.toml'
  Updating 'C:\Users\Andrew\.julia\environments\v1.12\Manifest.toml'
  [a65dc6b1] + Xorg_libpciaccess_jll v0.18.1+0
  [8e53e030] + libdrm_jll v2.4.125+1
  [9a156e7d] + libva_jll v2.23.0+0
  Precompiling packages finished.
  3 dependencies successfully precompiled in 30 seconds. 360 already precompiled.

julia> using DrWatson

julia> initialize_project("project"; authors="Ваше Имя", git=false)
```

Рисунок 1: Инициализация проекта

Устанавливаем все необходимые пакеты (Agents)

```
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> julia
```



Documentation: <https://docs.julialang.org>

Type "?" for help, "]??" for Pkg help.

Version 1.12.0 (2025-10-07)

Official <https://julialang.org> release

```
(@v1.12) pkg> add Agents
```

```
Resolving package versions...
```

```
Installed RealDot — v0.1.0
```

```
Installed OnlineStatsBase — v1.7.1
```

```
Installed LightSumTypes — v5.2.2
```

```
Cloning [94ee1d12-ae83-5a48-8b1c-48b8ff168ae0] Quaternions from https://github.com/JuliaGeometry/Quaternions.jl.git
```

```
Installed Quaternions — v0.7.7
```

```
Cloning [864edb3b-99cc-5e75-8d2d-829cb0a9cfe8] DataStructures from https://github.com/JuliaCollections/DataStructures.jl.git
```

```
Installed DataStructures — v0.18.22
```

```
Cloning [86223c79-3864-5bf0-83f7-82e725a168b6] Graphs from https://github.com/JuliaGraphs/Graphs.jl.git
```

```
Installed Graphs — v1.13.1
```

```
Cloning [46ada45e-f475-11e8-01d0-f70cc89e6671] Agents from https://github.com/JuliaDynamics/Agents.jl.git
```

```
Installed Agents — v7.0.0
```

```
Cloning [92933f4c-e287-5a05-a399-4b506db050ca] ProgressMeter from https://github.com/timholly/ProgressMeter.jl.git
```

```
Installed ProgressMeter — v1.11.0
```

```
Cloning [fff63dad9-3335-55d8-95ec-f8139d39e468] StreamSampling from https://github.com/JuliaDynamics/StreamSampling.jl.git
```

```
Installed StreamSampling — v0.7.6
```

```
Cloning [6038ab10-8711-5258-84ad-4b1120ba62dc] Rotations from https://github.com/JuliaGeometry/Rotations.jl.git
```

```
Installed Rotations — v1.7.1
```

Рисунок 2: Установка пакетов Agents

Устанавливаем дополнительные пакеты для работы с агентным моделированием

```
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> julia
```

[illegible]

```
julia> pkg
ERROR: UndefVarError: `pkg` not defined in `Main`
Suggestion: check for spelling errors or missing imports.
```

```
(@v1.12) pkg> add StatsBase
Resolving package versions...
Updating 'C:\Users\Andrew\.julia\environments\v1.12\Project.toml'
[2913bbd2] + StatsBase v0.34.10
Manifest No packages added to or removed from 'C:\Users\Andrew\.julia\environments\v1.12\Manifest.toml'
```

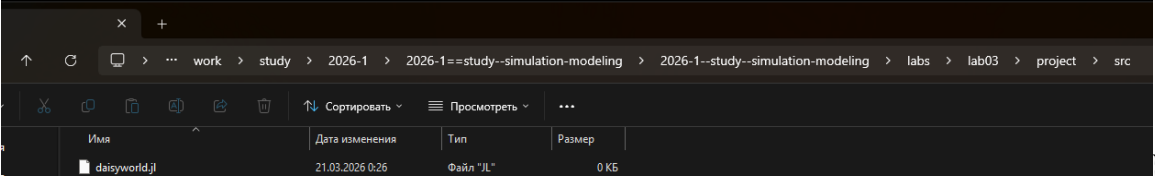
```
(@v1.12) pkg> ^C
```

```
julia> exit()
```

Рисунок 3: Загрузка доп. пакетов

Создаем файл в папке src, в котором мы определим тип агента и функции шага модели

```
Andrew@DESKTOP-3U5UIR8 MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
$ touch src/daisyworld.jl
```



The screenshot shows a file explorer window with the following directory path: `work > study > 2026-1 > 2026-1==study--simulation-modeling > 2026-1--study--simulation-modeling > labs > lab03 > project > src`. Below the path, there is a table listing the files in the current directory.

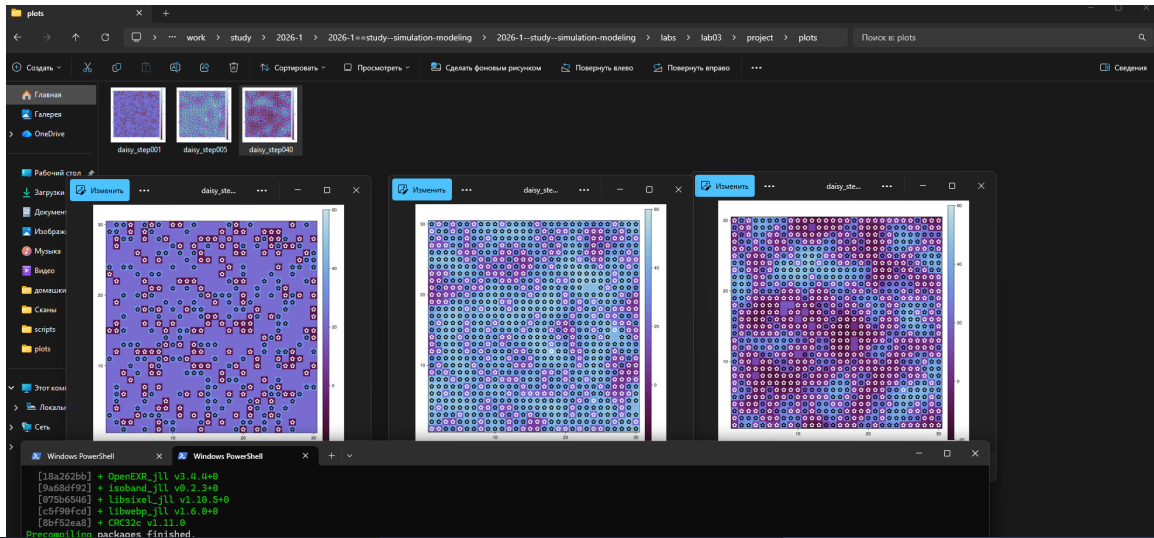
Имя	Дата изменения	Тип	Размер
daisyworld.jl	21.03.2026 0:26	Файл "JL"	0 КБ

Создаем исполняемый файл в папке scripts и прописываем в нем код

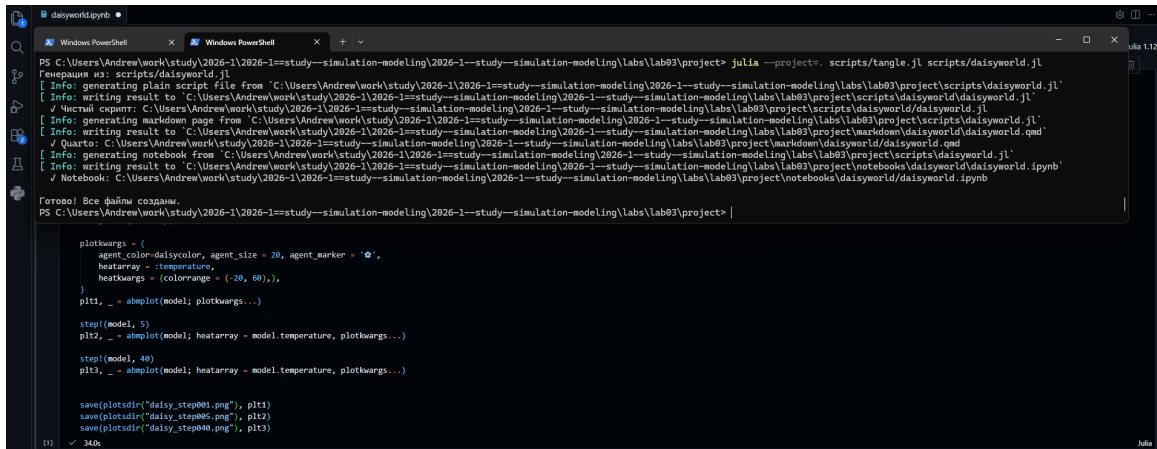
```
Andrew@DESKTOP-3U5UIRB MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/1  
$ touch scripts/daisyworld.jl
```

```
daisyworld.jl  
Файл  Изменить  Просмотр  
using DataFrames  
using Plots  
  
include(scrdir("daisyworld.jl"))  
  
using CairoMakie  
model = daisyworld()  
  
daisycolor(a::Daisy) = a.breed  
  
plotkwargs = (  
    agent_color=daisycolor, agent_size = 20, agent_marker = '✿',  
    heatarray = :temperature,  
    heatkwargs = (colorrange = (-20, 60)),  
)  
plt1, _ = abmplot(model; plotkwargs...)  
  
step!(model, 5)  
plt2, _ = abmplot(model; heatarray = model.temperature, plotkwargs...)  
  
step!(model, 40)  
plt3, _ = abmplot(model; heatarray = model.temperature, plotkwargs...)  
  
save(plt1, "daisy_step001.png")  
save(plt2, "daisy_step005.png")  
save(plt3, "daisy_step040.png")
```

Запускаем файл и анализируем тепловую карту, которая будет представлять собой температуру поверхности модели. Маргаритки будут отображаться черно-белыми в соответствии с их видом.



Преобразовываем код в произвольные стили, открываем jupyter файл и убеждаемся в правильной компиляции.



```
daisyworld.ipynb
Windows PowerShell
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> julia --project=. scripts/tangle.jl scripts/daisyworld.jl
Генерация из: scripts/daisyworld.jl
[ Info: generating plain script file from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld.jl'
[ Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld\daisyworld.jl'
√ Чистый скрипт: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld\daisyworld.jl
[ Info: generating markdown page from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld.jl'
[ Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\markdown\daisyworld\daisyworld.qmd'
√ Quarto: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\markdown\daisyworld\daisyworld.qmd
[ Info: generating notebook from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld.jl'
[ Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\notebooks\daisyworld\daisyworld.ipynb'
√ Notebook: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\notebooks\daisyworld\daisyworld.ipynb

Готово! Все файлы созданы.
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> |

plotkwargs = (
    agent_color=daisycolor, agent_size = 20, agent_marker = '☆',
    heatmap = :temperature,
    heatmapkwargs = (colorrange = (-20, 60).),
)
plt1, _ = abmplot(model; plotkwargs...)

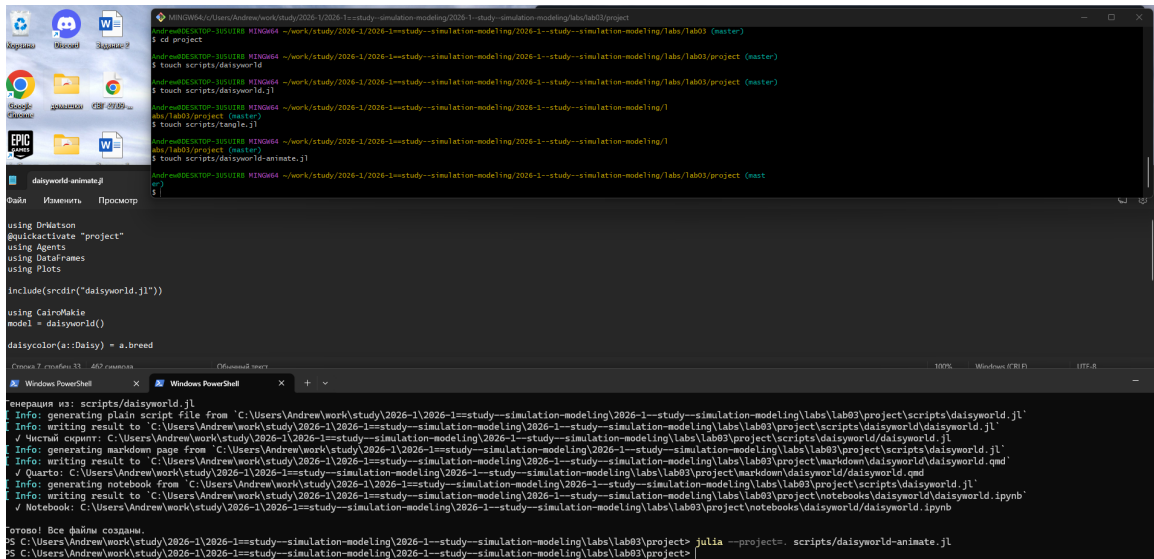
step!(model, 5)
plt2, _ = abmplot(model; heatmap = model.temperature, plotkwargs...)

step!(model, 40)
plt3, _ = abmplot(model; heatmap = model.temperature, plotkwargs...)

save(plotsdir("daisy_step001.png"), plt1)
save(plotsdir("daisy_step005.png"), plt2)
save(plotsdir("daisy_step040.png"), plt3)
```

Рисунок 6: Код из jupyter notebook

Создаем файл с анимацией, прописываем в нем код и запускаем.



The screenshot shows a Windows desktop environment. On the left, a file explorer window displays the contents of a folder named 'daisyworld-animate.jl'. The files listed include 'using DrWatson', '@quickactivate "project"', 'using Agents', 'using DataFrames', 'using Plots', 'include(srcdir("daisyworld.jl"))', 'using CairoMakie', 'model = daisyworld()', and 'daisycolor(a::Daisy) = a.breed'. On the right, a terminal window shows the execution of commands to create and run the project. The terminal output indicates the generation of a plain script file, a notebook, and a markdown page, as well as the writing of the result to the respective files.

```
MINGW64~/Users/Andrew/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project
Andrew@DESKTOP-3U5UIRB MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03 (master)
$ cd project
Andrew@DESKTOP-3U5UIRB MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
$ touch scripts/daisyworld
Andrew@DESKTOP-3U5UIRB MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
$ touch scripts/daisyworld.jl
Andrew@DESKTOP-3U5UIRB MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
$ touch scripts/tangle.jl
Andrew@DESKTOP-3U5UIRB MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
$ touch scripts/daisyworld-animate.jl
Andrew@DESKTOP-3U5UIRB MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
$
```

Строка 7, столбец 33... 462 символов Обычный текст 100% Windows (CRLF) UTF-8

Windows PowerShell x Windows PowerShell x + v

```
Генерация из: scripts/daisyworld.jl
[ Info: generating plain script file from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld.jl'
[ Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld\daisyworld.jl'
✓ Чистый скрипт: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld\daisyworld.jl
[ Info: generating markdown page from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld.jl'
[ Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\markdown\daisyworld\daisyworld.qmd'
✓ Quarto: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\markdown\daisyworld\daisyworld.qmd
[ Info: generating notebook from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld.jl'
[ Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\notebooks\daisyworld\daisyworld.ipynb'
✓ Notebook: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\notebooks\daisyworld\daisyworld.ipynb

Готово! Все файлы созданы.
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> julia --project=. scripts/daisyworld-animate.jl
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> |
```

Результат можно увидеть в папке plots или по ссылке в rutube:

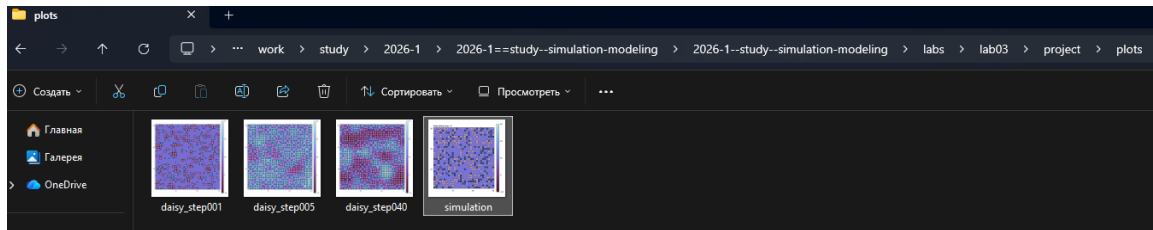


Рисунок 8: Результат

<https://rutube.ru/video/fc59731ceb08176e1f6a8170228c0836/>

Создаем, заполняем и запускаем файл, в котором наглядно показана динамика числа маргариток

```
MINGW64~/Users/Andrew/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project
andrew@DESKTOP-JUSUIR8 MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
$ touch scripts/tangle.jl

andrew@DESKTOP-JUSUIR8 MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
$ touch scripts/daisyworld-animate.jl

andrew@DESKTOP-JUSUIR8 MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
$ touch scripts/daisyworld-count.jl

andrew@DESKTOP-JUSUIR8 MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
$
```

daisyworld-count.jl

Файл

Изменить

Просмотр

```
using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots

include(srcdir("daisyworld.jl"))

using CairoMakie

black(a) = a.breed == :black
white(a) = a.breed == :white
```

Строка 9, столбец 1 731 символ

Обычный текст

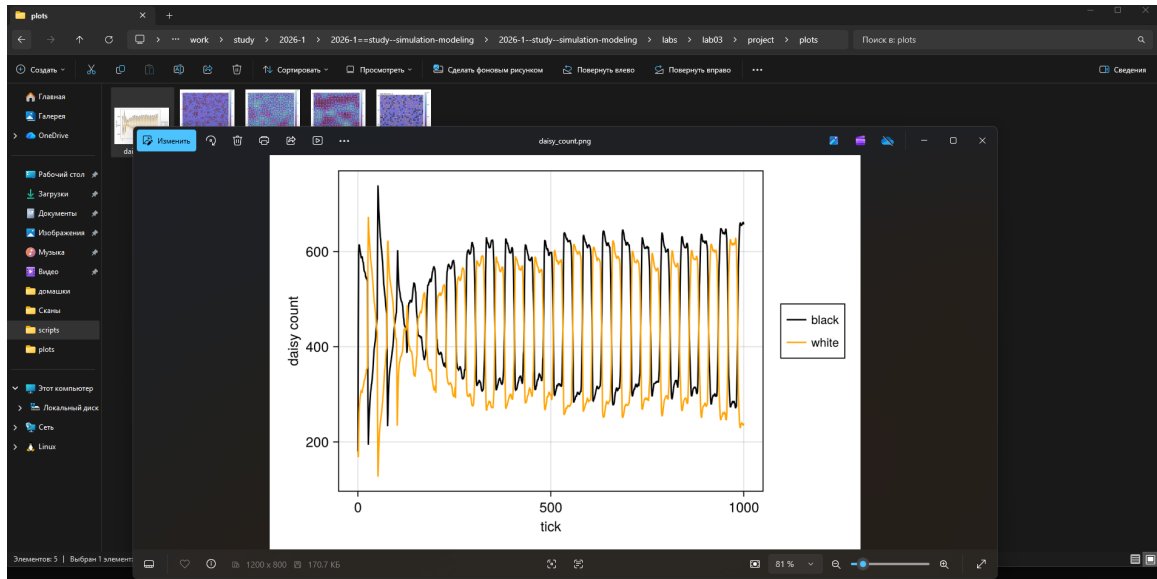
100% Windows (CRLF) UTF-8

```
Windows PowerShell
Windows PowerShell

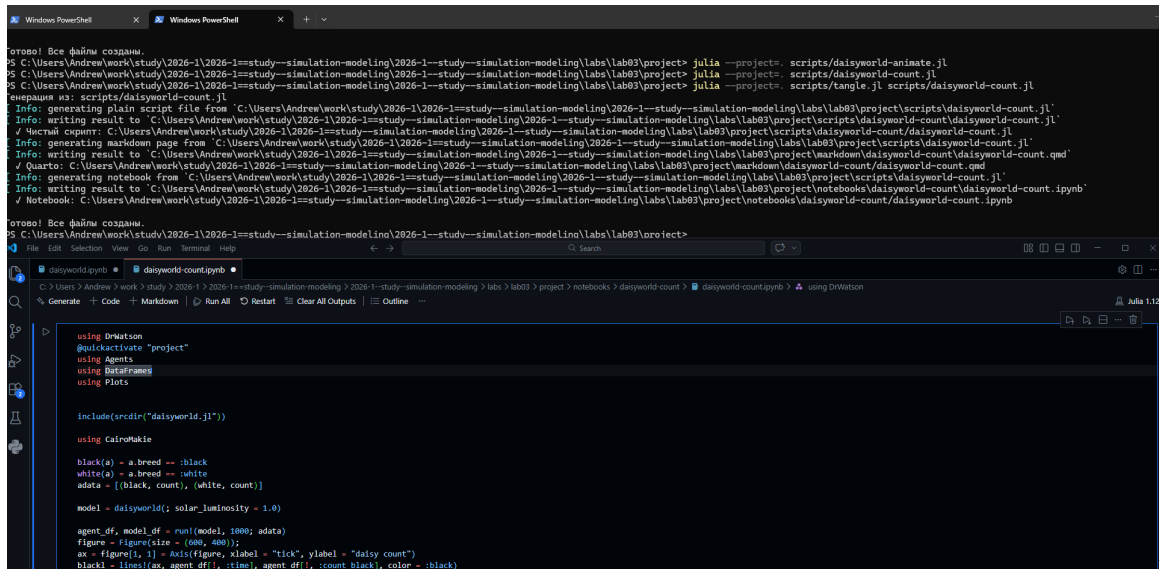
Генерация из: scripts/daisyworld.jl
Info: generating plain script file from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld.jl'
Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld\daisyworld.jl'
✓ Чистый скрипт: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld\daisyworld.jl
Info: generating markdown page from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld.jl'
Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\markdown\daisyworld\daisyworld.qmd'
✓ Quarto: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\markdown\daisyworld\daisyworld.qmd
Info: generating notebook from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld.jl'
Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\notebooks\daisyworld\daisyworld.ipynb'
✓ Notebook: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\notebooks\daisyworld\daisyworld.ipynb

Готово! Все файлы созданы.
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> julia --project=. scripts/daisyworld-animate.jl
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> julia --project=. scripts/daisyworld-count.jl
```

Наглядно видим результат запуска файла.



Преобразовываем код в произвольные стили, открываем jupyter файл и убеждаемся в правильной компиляции.



```
Готово! Все файлы созданы.
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> julia --project=. scripts\daisyworld-animate.jl
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> julia --project=. scripts\daisyworld-count.jl
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> julia --project=. scripts\tangle.jl scripts\daisyworld-count.jl
Генерация из: scripts\daisyworld-count.jl
Info: generating plain script file from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-count.jl'
Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-count\daisyworld-count.jl'
✓ Чистый скрипт: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-count\daisyworld-count.jl
Info: generating markdown page from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-count.jl'
Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\markdown\daisyworld-count\daisyworld-count.qmd'
✓ Quarto: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\markdown\daisyworld-count\daisyworld-count.qmd
Info: generating notebook from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-count.jl'
Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\notebooks\daisyworld-count\daisyworld-count.ipynb'
✓ Notebook: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\notebooks\daisyworld-count\daisyworld-count.ipynb

Готово! Все файлы созданы.
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project>
```

```
File Edit Selection View Go Run Terminal Help
← → Search
daisyworld.ipynb daisyworld-count.ipynb
C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\notebooks\daisyworld-count> daisyworld-count.ipynb using DrWatson
Generate + Code + Markdown Run All Restart Clear All Outputs Outline
Julia 1.12

using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots

include(sourcedir("daisyworld.jl"))

using CairoMakie

black(a) = a.breed == :black
white(a) = a.breed == :white
adata = [(black, count), (white, count)]

model = daisyworld(; solar_luminosity = 1.0)

agent_df, model_df = run!(model, 1000; adata)
figure = Figure(size = (600, 400));
ax = figure[1, 1] = Axis(figure, xlabel = "tick", ylabel = "daisy count")
blackl = lines!(ax, agent_df[:, :time], agent_df[:, :count_black], color = :black)
```

Создаем, заполняем и запускаем файл, который построит комплексный график изменения числа маргариток, температуры, альбедо в зависимости от модельного времени

```
MINGW64 C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project
andrew@DESKTOP-3USUE18 MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/l
ls /lab03/project (master)
touch scripts/daisyworld-animate.jl
andrew@DESKTOP-3USUE18 MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (mast
*)
touch scripts/daisyworld-count.jl
andrew@DESKTOP-3USUE18 MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
touch scripts/daisyworld-luminosity.jl
andrew@DESKTOP-3USUE18 MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)

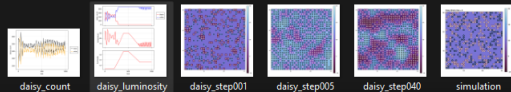
Файл Изменить Просмотр

figure = CairoMakie.Figure(size = (600, 600));
ax1 = Figure[1, 1] = Axis{figure, ylabel = "daisy count"}
black1 = lines!(ax1, agent_df[:, :time], agent_df[:, :count_black], color = :red)
white1 = lines!(ax1, agent_df[:, :time], agent_df[:, :count_white], color = :blue)
figure[1, 2] = Legend{figure, [black1, white1], ["black", "white"]}

ax2 = Figure[2, 1] = Axis{figure, ylabel = "temperature"}
ax3 = Figure[3, 1] = Axis{figure, xlabel = "tick", ylabel = "luminosity"}
lines!(ax2, model_df[:, :time], model_df[:, :temperature], color = :red)
lines!(ax3, model_df[:, :time], model_df[:, :solar_luminosity], color = :red)
for ax in (ax1, ax2); ax.xtickslabelsvisible = false; end
figure

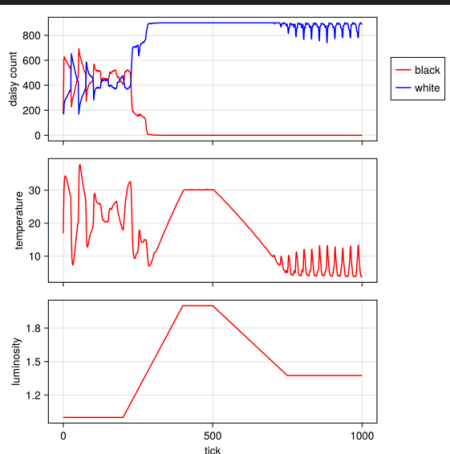
Windows PowerShell
Готово! Все файлы созданы.
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> julia --project=. scripts/daisyworld-animate.jl
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> julia --project=. scripts/daisyworld-count.jl
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> julia --project=. scripts/tangle.jl scripts/daisyworld-count.jl
Генерация из: scripts/daisyworld-count.jl
[ Info: generating plain script file from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-count.jl'
[ Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-count\daisyworld-count.jl'
✓ Чистый скрипт: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-count\daisyworld-count.jl
[ Info: generating markdown page from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-count.jl'
[ Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\markdown\daisyworld-count\daisyworld-count.qmd'
✓ Quarto: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\markdown\daisyworld-count\daisyworld-count.qmd
[ Info: generating notebook from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-count.jl'
[ Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\notebooks\daisyworld-count\daisyworld-count.ipynb'
✓ Notebook: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\notebooks\daisyworld-count\daisyworld-count.ipynb
Готово! Все файлы созданы.
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> julia --project=. scripts/daisyworld-luminosity.jl
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project>
```

Заходим в папку plots и открываем наши графики, анализируем поведение модели

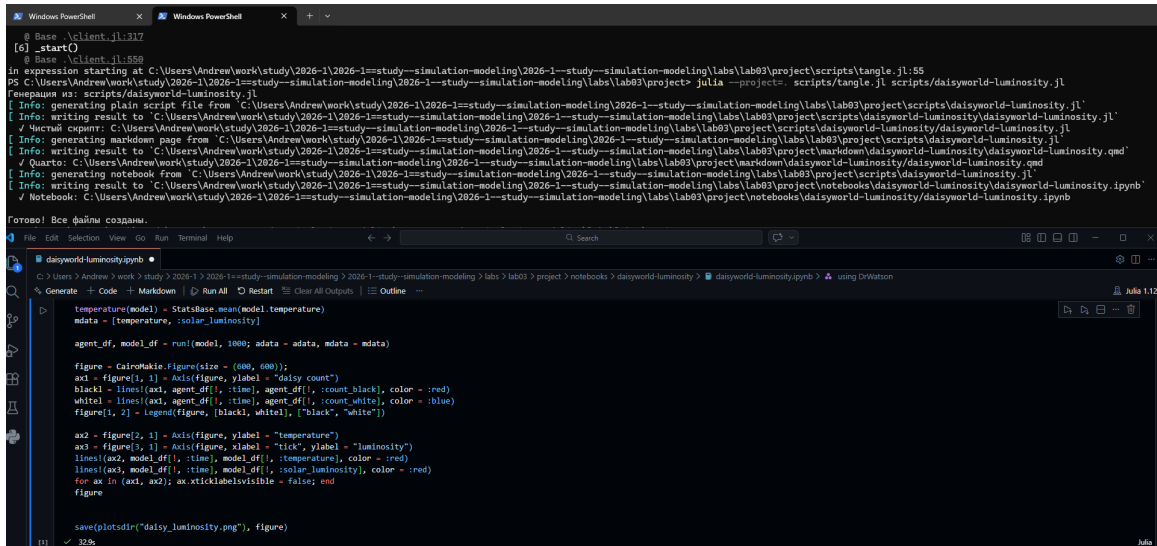


Изменить

daisy_luminosity.png



Преобразовываем код в произвольные стили, открываем jupyter файл и убеждаемся в правильной компиляции.



```
@ Base .\client.jl:317
[6] _start()
@ Base .\client.jl:550
in expression starting at C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\tangle.jl:55
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> julia --project=. scripts\tangle.jl scripts\daisyworld-luminosity.jl
Генерация из: scripts\daisyworld-luminosity.jl
[ Info: generating plain script file from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-luminosity.jl'
[ Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-luminosity.jl'
✓ Чистый скрипт: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-luminosity\daisyworld-luminosity.jl
[ Info: generating markdown page from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-luminosity.jl'
[ Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\markdown\daisyworld-luminosity\daisyworld-luminosity.qmd'
✓ Quarto: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\markdown\daisyworld-luminosity\daisyworld-luminosity.qmd
[ Info: generating notebook from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-luminosity.jl'
[ Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\notebooks\daisyworld-luminosity\daisyworld-luminosity.ipynb'
✓ Notebook: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\notebooks\daisyworld-luminosity\daisyworld-luminosity.ipynb

Готово! Все файлы созданы.
```

```
temperature(model) = StatsBase.mean(model.temperature)
mdata = [temperature, :solar_luminosity]

agent_df, model_df = run!(model, 1000; adata = adata, mdata = mdata)

figure = CairoMakie.Figure(size = (600, 600));
ax1 = figure[1, 1] = Axis{figure, ylabel = "daisy count"}
blackl = lines!(ax1, agent_df[:, :time], agent_df[:, :count_black], color = :red)
whitel = lines!(ax1, agent_df[:, :time], agent_df[:, :count_white], color = :blue)
figure[1, 2] = Legend(figure, [blackl, whitel], ["black", "white"])

ax2 = figure[2, 1] = Axis{figure, ylabel = "temperature"}
ax3 = figure[3, 1] = Axis{figure, xlabel = "tick", ylabel = "luminosity"}
lines!(ax2, model_df[:, :time], model_df[:, :temperature], color = :red)
lines!(ax3, model_df[:, :time], model_df[:, :solar_luminosity], color = :red)
for ax in (ax1, ax2); ax.xtickslabelsvisible = false; end
figure

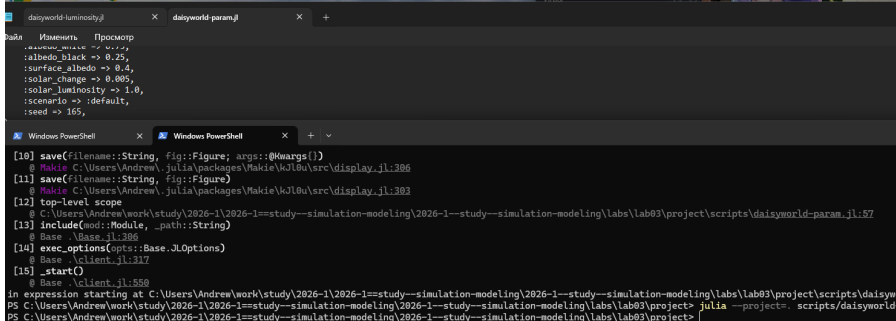
save(plotsdir("daisy_luminosity.png"), figure)
```

Мы можем расширить базовую визуализацию за счет параметров. Для этого создаем новый файл, в котором прописываем параметры и на выходе получаем несколько троек графиков для каждого набора параметров

```
MINGW64/c/Users/Andrew/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project
ndr@DESKTOP-SUSUIR8 MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
$ touch scripts/daisyworld-luminosity.jl

ndr@DESKTOP-SUSUIR8 MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
$ touch scripts/daisyworld_param.jl

ndr@DESKTOP-SUSUIR8 MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
$
```

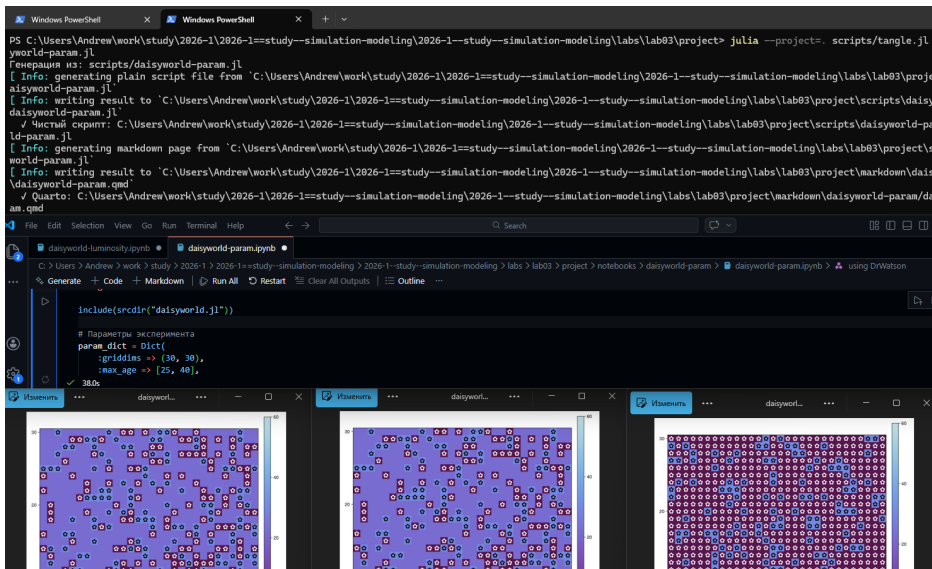


```
File  daisyworld-luminosity.jl  daisyworld-param.jl  +
File  Изменить  Просмотр
.albedo_white -> 0.75,
:albedo_black -> 0.25,
:surface_albedo -> 0.4,
:solar_change -> 0.005,
:solar_luminosity -> 1.0,
:scenario -> :default,
:seed -> 165,

Windows PowerShell  Windows PowerShell  +
[10] save(filename::String, fig::Figure; args::@Kwargs{})
@ Makie C:\Users\Andrew\.julia\packages\Makie\kJL0u\src\display.jl:306
[11] save(filename::String, fig::Figure)
@ Makie C:\Users\Andrew\.julia\packages\Makie\kJL0u\src\display.jl:303
[12] top-level scope
@ C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-param.jl:57
[13] include(mod::Module, _path::String)
@ Base .\Base.jl:306
[14] exec_options(opts::Base.JLOptions)
@ Base .\client.jl:317
[15] _start()
@ Base .\client.jl:550
in expression starting at C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld.jl:57
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> julia --project=. scripts/daisyworld.jl
PS C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project> |
```



Преобразовываем код в произвольные стили, открываем jupyter файл и убеждаемся в правильной компиляции.



Создаем файл, в котором зависимо от заданного набора параметров можно построить график динамика числа маргаритов в зависимости от времени, заполняем и запускаем его

```
MINGW64~/C:/Users/Andrew/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project
ndrew@DESKTOP-JUSUIR8 MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
touch scripts/daisyworld__param.jl

ndrew@DESKTOP-JUSUIR8 MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
touch scripts/daisyworld-count__param.jl

ndrew@DESKTOP-JUSUIR8 MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
```

```
Без имени daisyworld-count-param.jl x +
Файл Изменить Просмотр
:solar_luminosity => 1.0,
:scenario => :default,
:seed => 165,
)

## Создаём список всех комбинаций
params_list = dict_list(param_dict)
counter = 0
for params in params_list

    model = daisyworld(;params...)

    agent_df, model_df = run!(model, 1000; adata)
    figure = Figure(size = (600, 400));
    ax = figure[1, 1] = Axis(figure, xlabel = "tick", ylabel = "daisy count")
    blackl = lines!(ax, agent_df[!, :time], agent_df[!, :count_black], color = :black)
    whitel = lines!(ax, agent_df[!, :time], agent_df[!, :count_white], color = :orange)
    Legend(figure[1, 2], [blackl, whitel], ["black", "white"], labelsz = 12)
    global counter
    counter += 1
    plt_name = "daisy-count_$(counter).png"

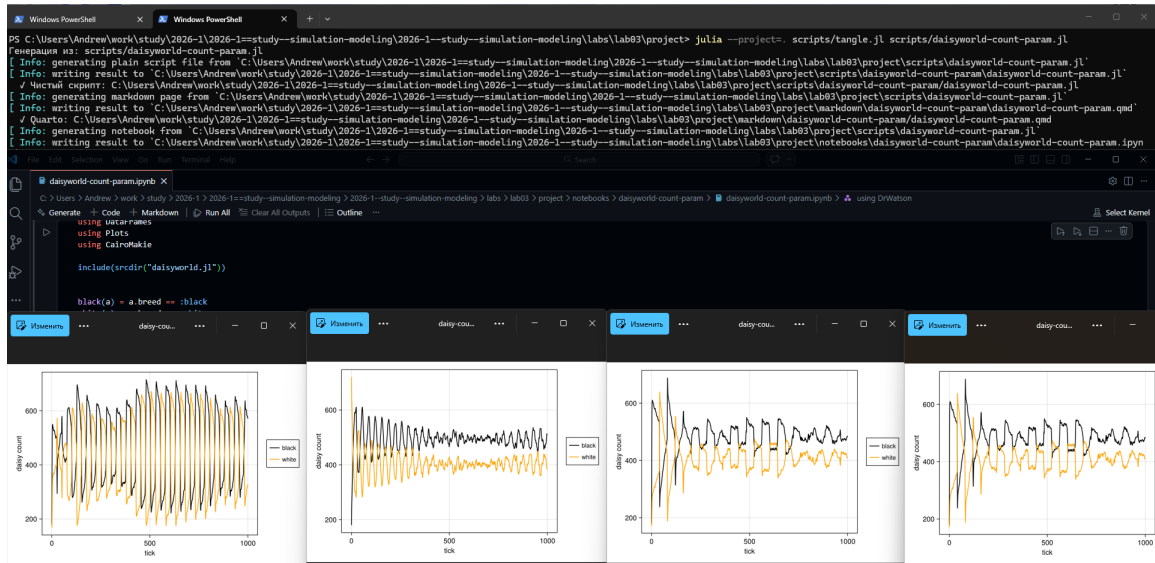
    save(plotsdir(plt_name), figure)

end
```

```
Windows PowerShell x Windows PowerShell x + v
counter += 1
plt_name = "daisy-count_$(counter).png" ".png"
# l — Expected 'end'

Stacktrace:
[1] top-level scope
@ C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-count-param.jl:45
[2] include(mod::Module, _path::String)
@ Base .\Base.jl:306
[3] exec_options(opts::Base.JLOptions)
```

Преобразовываем код в произвольные стили, открываем jupyter файл. анализируем получившиеся графики,зависящие от наших заданных параметров



Создаем, заполняем и запускаем файл, в котором можно построить комплексные графики, задав произвольные параметры

```
MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project
ndrew@DESKTOP-SUSUER MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
touch scripts/daisyworld-count-param.jl

ndrew@DESKTOP-SUSUER MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)
touch scripts/daisyworld-luminosity-param.jl

ndrew@DESKTOP-SUSUER MINGW64 ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project (master)

Без имени daisyworld-count-param.jl daisyworld-luminosity-param.jl +
Файл Изменить Просмотр
using DataFrames
using Plots
using CairoMakie

include(srcdir("daisyworld.jl"))

black(a) = a.breed == :black
white(a) = a.breed == :white
adata = [(black, count), (white, count)]

## Параметры эксперимента
param_dict = Dict{
  :griddims => (30, 30),
  :max_age => [25, 40],
  :init_white => [0.2, 0.8],
  :init_black => 0.2,
  :albedo_white => 0.75,
  :albedo_black => 0.25,
  :surface_albedo => 0.4,
  :solar_change => 0.005,
  :solar_luminosity => 1.0,
  :scenario => :ramp,
  :seed => 165,
}

Строка 23, столбец 28 1738 символов Обычный текст 100% Windows (CRLF) UTF-8
Windows PowerShell Windows PowerShell

ам.jl'
✓ Чистый скрипт: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-luminosity-param\daisyworld-luminosity-param.jl
[ Info: generating markdown page from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-luminosity-param.jl'
[ Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\markdown\daisyworld-luminosity-param\daisyworld-luminosity-param.qmd'
✓ Quarto: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\markdown\daisyworld-luminosity-param\daisyworld-luminosity-param.qmd
[ Info: generating notebook from 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\scripts\daisyworld-luminosity-param.jl'
[ Info: writing result to 'C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\notebooks\daisyworld-luminosity-param\daisyworld-luminosity-param.ipynb'
✓ Notebook: C:\Users\Andrew\work\study\2026-1\2026-1==study--simulation-modeling\2026-1--study--simulation-modeling\labs\lab03\project\notebooks\daisyworld-luminosity-param\daisyworld-luminosity-param.ipynb

Сотворил все файлы созданы
```

Преобразовываем код в произвольные стили, открываем jupyter файл. Анализируем получившиеся графики,зависящие от наших заданных параметров



В данной работе мы разобрались с агентным моделированием и реализовали его на практике.